COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithm	**SECTION:**	CPE 21S1
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ACTIVITY NO. 2 **Hands-on Activity 1.1 Basic C++ Programming**

1. PROCEDURE OUTPUT

Sections:	Answer:
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`int a, b, c, d;`

Class Declaration and Method Definition Section	`int getSum(int a, int b);`
	`void compare(int a, int b);`
	`void logic(bool c, bool d);`

Main Function

```
int main(){
    int choice;

    do{
        std::cout<<"Input first num: ";
        std::cin>>a;
        std::cout<<"Input second num: ";
        std::cin>>b;
        std::cout<<"Input a boolean value: ";
        std::cin>>c;
        std::cout<<"Input a second boolean value: ";
        std::cin>>d;

        std::cout<<"\nThe sum of the two numbers is " << getSum(a, b) << std::endl; //display sum
        compare(a, b); //display whether a is greater than b
        logic(c,d); //displays all logical operations

        std::cout<<"\nPress 1 to redo, otherwise press any num: ";
        std::cin>>choice;

    }while(choice == 1); //to redo
}
```

Method Definition

```
int getSum(int a, int b){
    int sum=0;
    sum = a + b;
    return sum;
}
```

```
void compare(int a, int b){
    if (a > b){
        std::cout<< a << " is greater than " << b;
    } else{
        std::cout<< a << " is not greater than " << b;
    }
    return;
}
```

```
void logic(bool c, bool d){
    std::cout<<"\n\n>>>LOGICAL OPERATION<<<"<<std::endl;
    std::cout<<"c && d = "<<(c && d)<<std::endl;
    std::cout<<"c || d = "<<(c || d)<<std::endl;
    std::cout<<"!c = "<<(!c)<<std::endl;
    std::cout<<"!d = "<<(!d);

    return;
}
```

Table 1-1. C++ Structure Code for Answer

```cpp
#include <iostream>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}
```

```cpp
int main() {
    // driver code
    Triangle set1(40, 30, 110);
    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }
    return 0;
}
```

```
The shape is a valid Triangle
```

Observation: I noticed that in this code, there is a slight error in the total angle part as it says less than or equal to 180 it is a triangle it should only be equal to as if the numbers that I set on the driver code is less than 180 it will still output that the shape is a valide triangle even though it is not.

Table 1-2. ILO B output observations and comments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a C++ program to swap the two numbers in different variables.

```cpp
#include <iostream>

//function prototype
void swappa(int &a, int &b);

int main() {
    int n1, n2;

    std::cout << "Enter first number: "; //asks user to input 1st num
    std::cin >> n1;
    std::cout << "Enter second number: ";//asks user to input 2nd num
    std::cin >> n2;

    std::cout << "\nBefore swapping: " << n1 << " and " << n2 << std::endl; //outputs the num before swap
    swappa(n1, n2); //calls for the function
    std::cout << "After swapping: " << n1 << " and " << n2 << std::endl; //outputs the swapped numbers

    return 0;
}
//function definition
void swappa(int &a, int &b) {
    int temp = a; //stores value of a in temp
    a = b; //assigns value of b to a
    b = temp; //assigns value of temp to b
}
```

```
Enter first number: 10
Enter second number: 20

Before swapping: 10 and 20
After swapping: 20 and 10
```

Table 2-1. Swap code and output

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```cpp
#include <iostream>

float kToF(float kelvin); //function prototype

int main() {
    float kelvin;

    std::cout << "Enter temp in Kelvin: "; //asks user to input temp in Kelvin
    std::cin >> kelvin;

    float fahren = kToF(kelvin); //calls for the function to convert K to F

    std::cout << kelvin << " K = " << fahren << " F" << std::endl; //outputs the converted temp

    return 0;
}

float kToF(float kelvin) { //function definition
    return (kelvin - 273.15) * 9.0 / 5.0 + 32; //converts K to F
}
```

```
Enter temp in Kelvin: 2112
2112 K = 3341.93 F
```

Table 2-2. Kelvin to Fahrenheit code and output

3. Create a C++ program that has a function that will calculate the distance between two points.

```cpp
#include <iostream>
#include <cmath>

float compD(float d1, float dd1, float d2, float dd2); //function prototype

int main() {
    float d1, dd1, d2, dd2;

    std::cout << "Enter d1 and dd1: "; //asks user to input d1 and dd1
    std::cin >> d1 >> dd1;
    std::cout << "Enter d2 and dd2: "; //asks user to input d2 and dd2
    std::cin >> d2 >> dd2;

    float distance = compD(d1, dd1, d2, dd2); //calls the function and store it in the variable distance

    std::cout << "The distance is: " << distance << std::endl; //outputs the distance

    return 0;
}

float compD(float d1, float dd1, float d2, float dd2) {
    return sqrt((d2-d1)*(d2-d1) + (dd2-dd1)*(dd2-dd1)); //calculates the distance
}
```

```
Enter d1 and dd1: 2
4
Enter d2 and dd2: 3
6
The distance is: 2.23607
```

Table 2-3. Distance code and output

4. Modify the code given in ILO B and add the following functions:
   a. A function to compute for the area of a triangle
   b. A function to compute for the perimeter of a triangle
   c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```cpp
#include <iostream>
#include <cmath>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();

    //NEWLY ADDED
    double compArea(double a, double b, double c); //func prot. for computing area
    double compPmeter(double a, double b, double c); //func prot. for computing perimeter
    std::string whataTriangle(); //func prot. for finding what kind of angle
};

//the original functions in here was transferred after the int main

int main() {
    // driver code
    Triangle set1(40, 30, 110);
    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    //NEW ADDITIONS
```

```
    std::cout << "Area: " << set1.compArea(6, 7, 11) << std::endl; //The 6, 7, 11 are the different sides to compute the area
    std::cout << "Perimeter: " << set1.compPmeter(6, 9, 11) << std::endl; //The 6, 9, 11 are the sides to compute perimeter
    std::cout << "What angle: " << set1.whataTriangle() << std::endl; //outputs the function for what kind of angle is the triangle

    return 0;
}

//The function definitions
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}
```

```
const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}

//NEW function definitions
double Triangle::compArea(double a, double b, double c) { //func def. for computing area
    double s = (a+b+c)/2;
    return sqrt(s*(s-a)*(s-b)*(s-c));
}

double Triangle::compPmeter(double a, double b, double c) { //func def. for computing perimeter
    return a + b + c;
}

std::string Triangle::whataTriangle() { //func def. for finding what kind of angle
    if (angleA<90 && angleB<90 && angleC<90) { //acute means less than 90, even if one is 90, it is not acute
        return "acute-angled";
    } else if (angleA>90 || angleB>90 || angleC>90) { //obtuse means greater than 90, if one is 90, it is not obtuse
        return "obtuse-angled";
    } else {
        return "others..."; //others are just others.. like right triangle and stuff
    }
}
```

```
The shape is a valid triangle.
Area: 18.9737
Perimeter: 26
What angle: obtuse-angled
```

Table 2-4. Triangle2 code and output

Conclusion

In this HOA, I relearned all my past lessons in CPE007 as well as learning new topics in ILO B which is to make a class with the private and public functions. I relearned many lessons like swapping variables just from switching the variable to another temporary variable and vice versa, as well as relearning all the forgotten syntaxes and what to do in logical operators like AND as well as OR. I also relearned to use more libraries like the cmath library which

lets me use the built in functions in it like square roots and exponents. I also relearned similar problems in CPE007 like the conversions of temperatures as well as areas and perimeters of a shape.

In ILO A, I made all 3 functions in 1 code and followed all the notes that stated that the driver program must call each function and that the definitions must be after the main function. So what I did first was that I made a function to get the sum of 2 numbers by asking first the user to input these 2 numbers and then computing it. Then I made a function to determine whether the first number is greater than or less than the second number. Lastly, I made a function to display all logic operandees like AND, OR, and NOT by first asking the user for 2 boolean numbers which is 1 and 0 (I forgot to put some verification so that the user can only do 1 and 0). Meanwhile in ILO B, I followed the guide step by step by first adding headers, then creating class w/ the private and public functions, then creating the function definitions then lastly, the int main.

In the supplementary, I first made a swapper code by using a temporary variable to store the first num, then stores the second num to first num, then stores the temporary variable to second num. Secondly, I made a code Kelvin to Fahreinheit, though I didn't put Fahrenheit to Kelvin, but I used a formula to first compute the 2 values that was asked in the the int main then the function is called then outputted. Next, it is similar to the temp code but with distance, I asked the user for 4 kinds of number to compute the distances then those will be computed in the function and the function will be called and outputted. Last of all is the code that I spent the most time with, so I first copy pasted the whole code of ILO B then added some codes within in, so I first added 3 more function prototypes in the public functions, then I made the definitions under the int main. So I first made the function that computes the Area, I used the cmath to be able to use the square root function, then I made the second function which computes the parameter which is simple enough by just adding, then the last of all is the function that determines what kind of angle it is, I used logic gates for it to make it more easier for me and to apply what I learned in ILO A, then I also used if else statements to keep comparing these numbers.

Overall, I think I did my mid well in this, as I lack the creativity to add more things as I forgotten to add more verifications and the like. I plan to improve on my creativity to add more things and to not make my code dull.

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Tools Used: **Claude**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of concepts as well as helping me to run and debug in VSCode but my VSCode doesn't run properly even with the help**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used claude to further explain to me concepts that I had forgotten as well as helped me in the analyzation for ILO B and concepts regarding public/ private. I used AI to help me run my VSCode but it was in vain as I cannot run my VSCode and just used my Codeblocks instead to run my program while getting a screenshot of my code in VSCode. It even suggested for me to download ubuntu or take down the security for the VSCode to work, but I just used Codeblocks because it was a hassle.**

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